Movement
Key Concepts

Continuous-Time Animal Movement Modeling

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Usha Kiran, 2024


Larry Li, 2016


David Courbit, 2019

Overview

Overview

1) Introduction to movement ecology
2) Projections
3) Measurements
4) Orientation
5) Stochastic processes
6) Defining movement

Overview

1) **Introduction to movement ecology**
2) Projections
3) Measurements
4) Orientation
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6) Defining movement

Introduction

Movement is a fundamental characteristic of life.

It dictates how individuals **acquire resources**, **evade predators**, **exchange genetic material**, and **respond to stressful environments**.

Mark Gocke

Ray F.

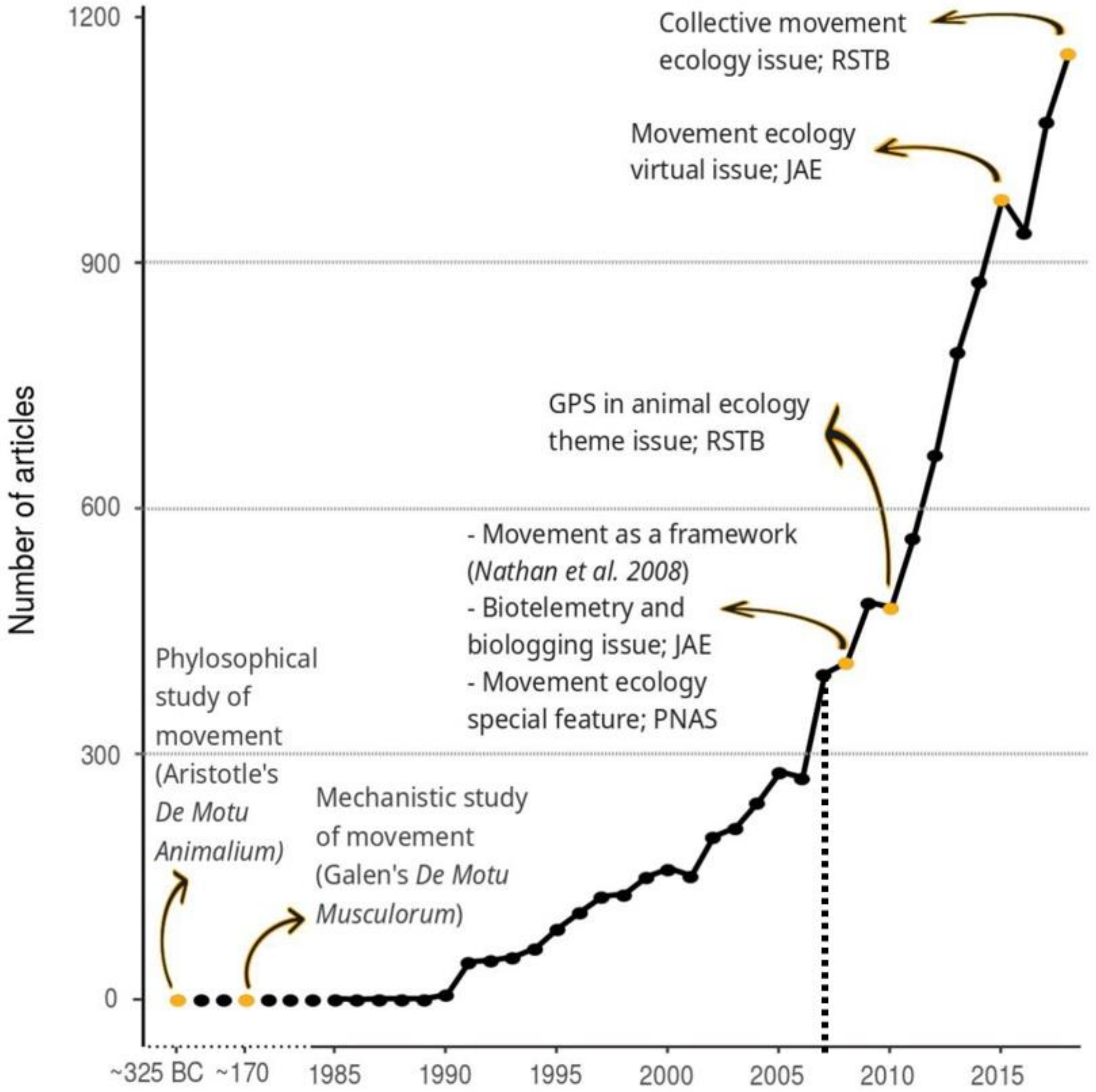
Grégoire Bouguereau

Cody Schroeder

Introduction

In 2008, "**movement ecology**" was distinguished from other ecological studies on behavior by its focus on the *movement of individuals*.

- Joo et al. (2022)



Introduction

Scientists have been systematically collecting animal movement data since the 1800s, employing ever-evolving technologies:

Data collection methods:

Fred Zwicky

1800s
Bird banding

1940s
Radar

Patricia Lim

1960s
Radio-telemetry

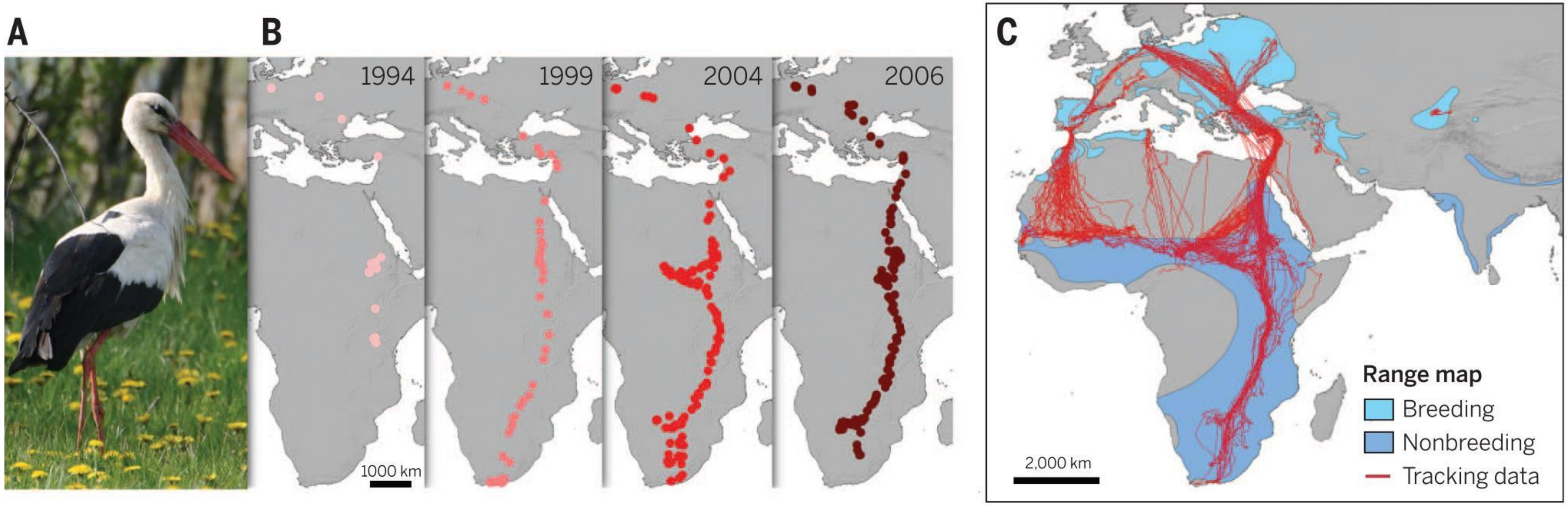
1990s
GPS technology

2000s
Nanotags, drones

Biologging technologies
(animal-carried devices that collect and/or transmit data)

Introduction

- Adapted from Kays et al. (2015)
  - A white stork (*Ciconia ciconia*) (A) was tagged with a GPS tracking device as a 3-year-old nonreproductive juvenile in Germany in 1994, and was tracked until her death in 2006 (B).

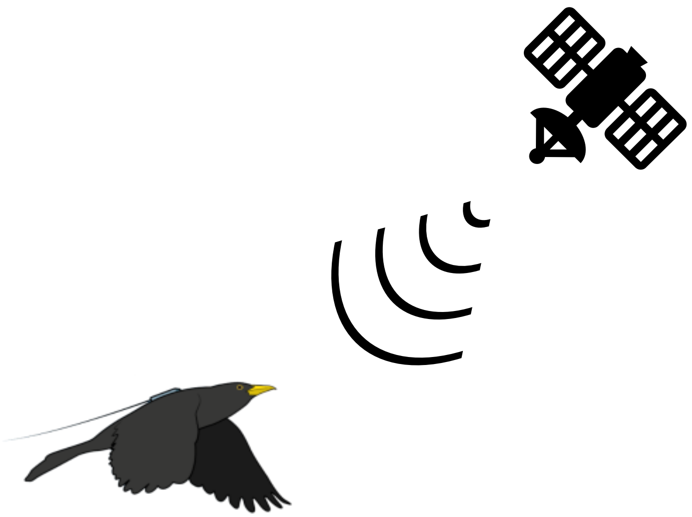


VHF tracking

- Cheaper per unit
- Labor intensive
  - Unless automated VHF systems

GPS tracking

- Expensive per unit
- Automated



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  - Also affected by environmental features & topography

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  - Or if environmental features obscure visibility of the sky (clouds)

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  - Sparse or irregular data

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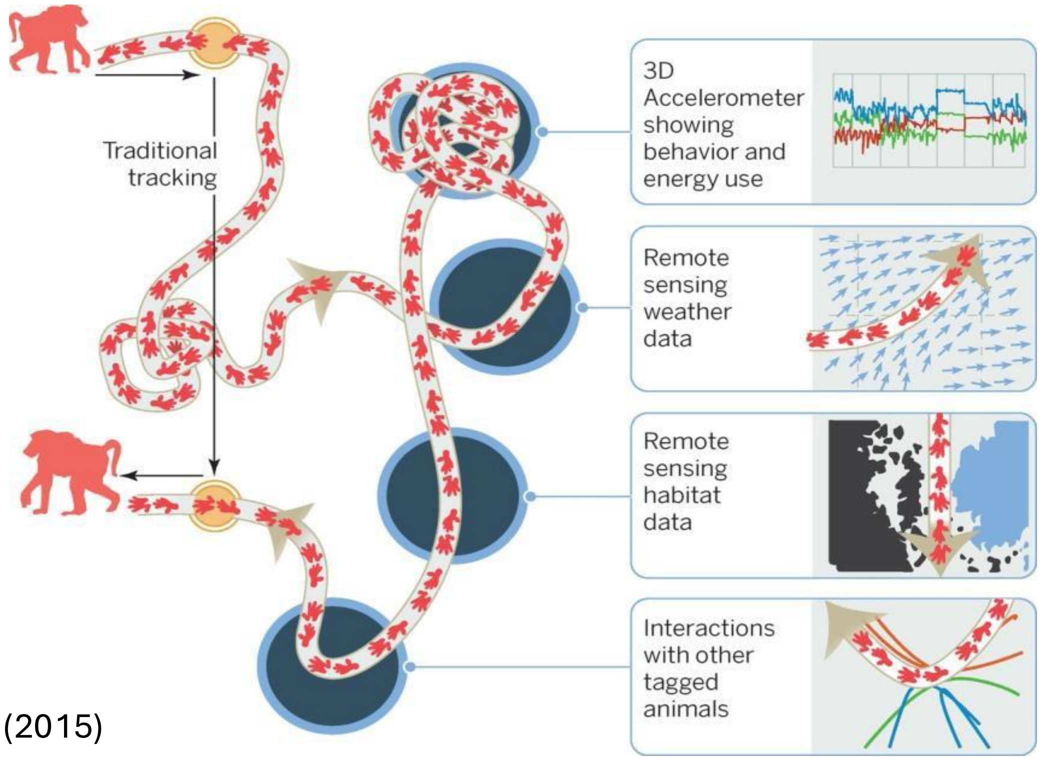
Tag placement may affect locomotion, thermoregulation, or increase risk of injuries or entanglement.

Introduction

Why do we study movement?

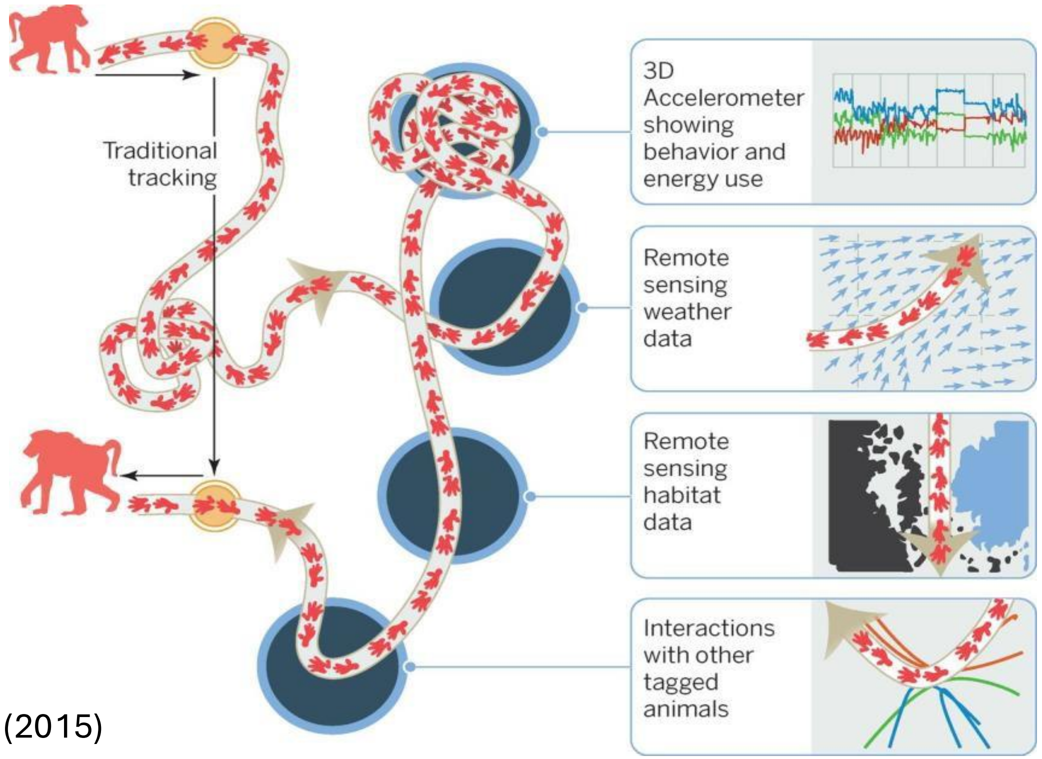


Kays et al. (2015)

Introduction

Why do we study movement?

- Broadly, we consider 3 fundamental questions:



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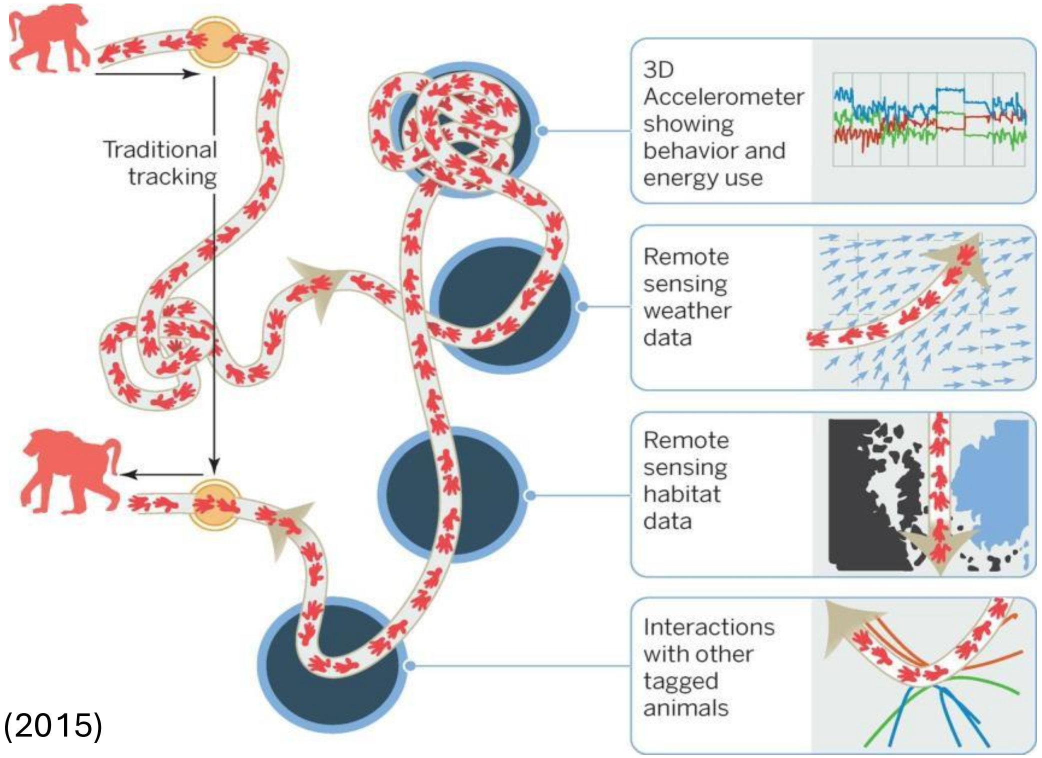

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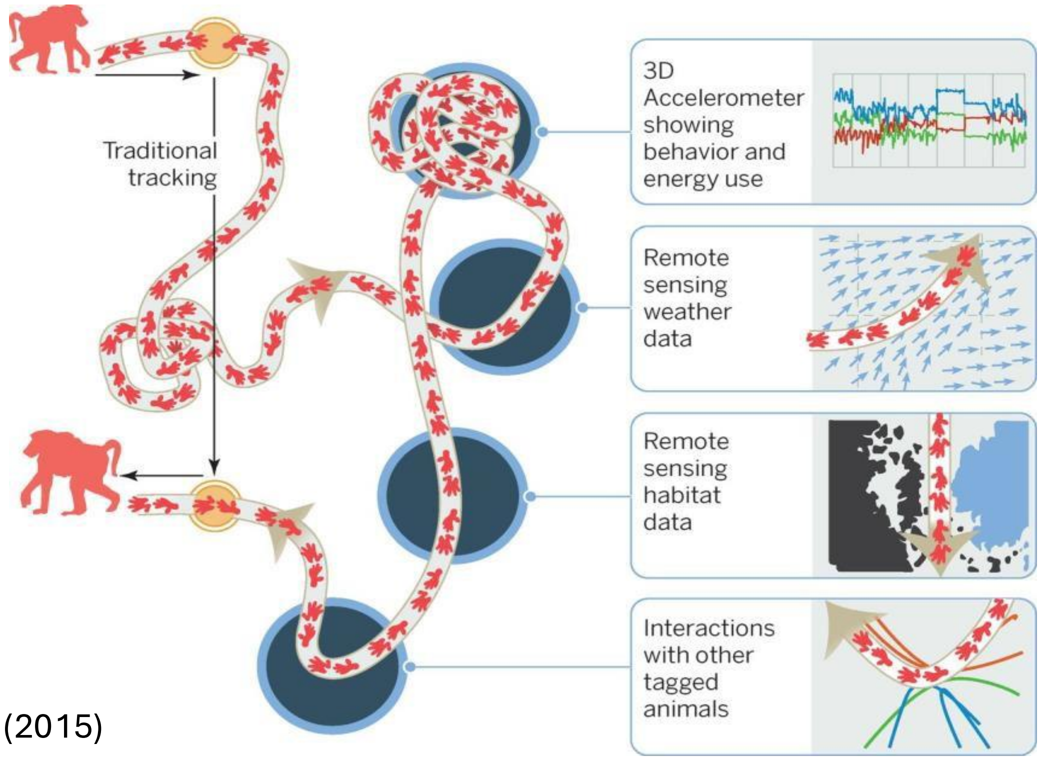
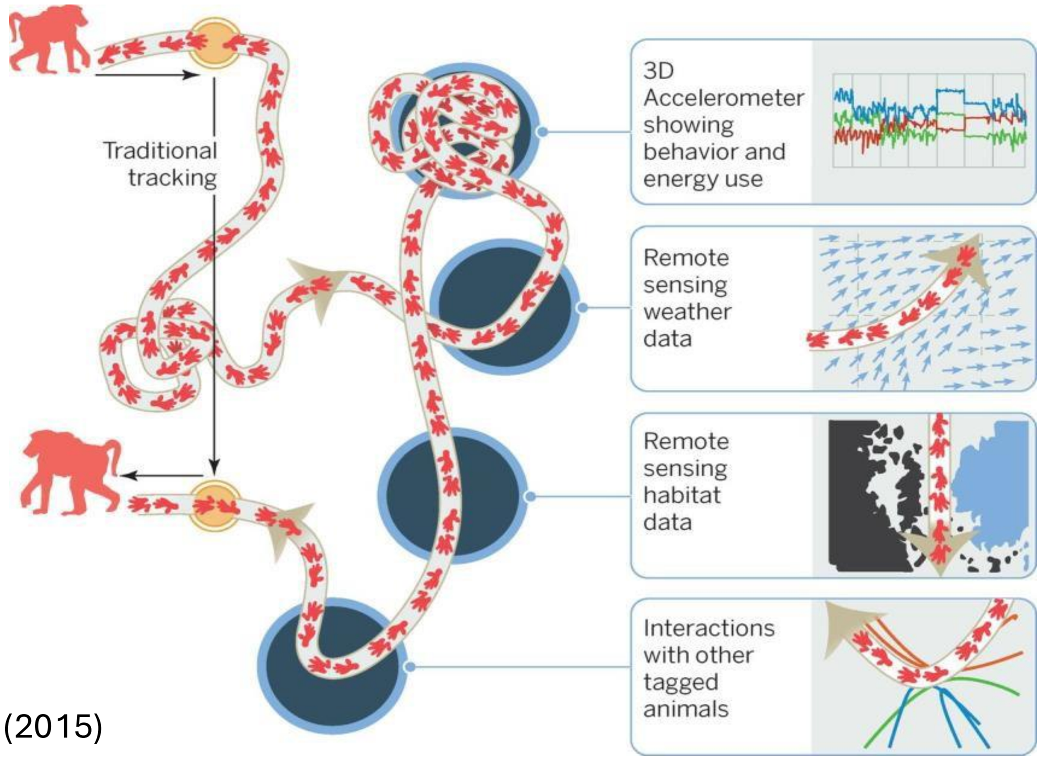


Kays et al. (2015)

Introduction

Why do we study movement?

- Broadly, we consider 3 fundamental questions:
  1) Why move?
  2) How to move?
  3) When and where to move?



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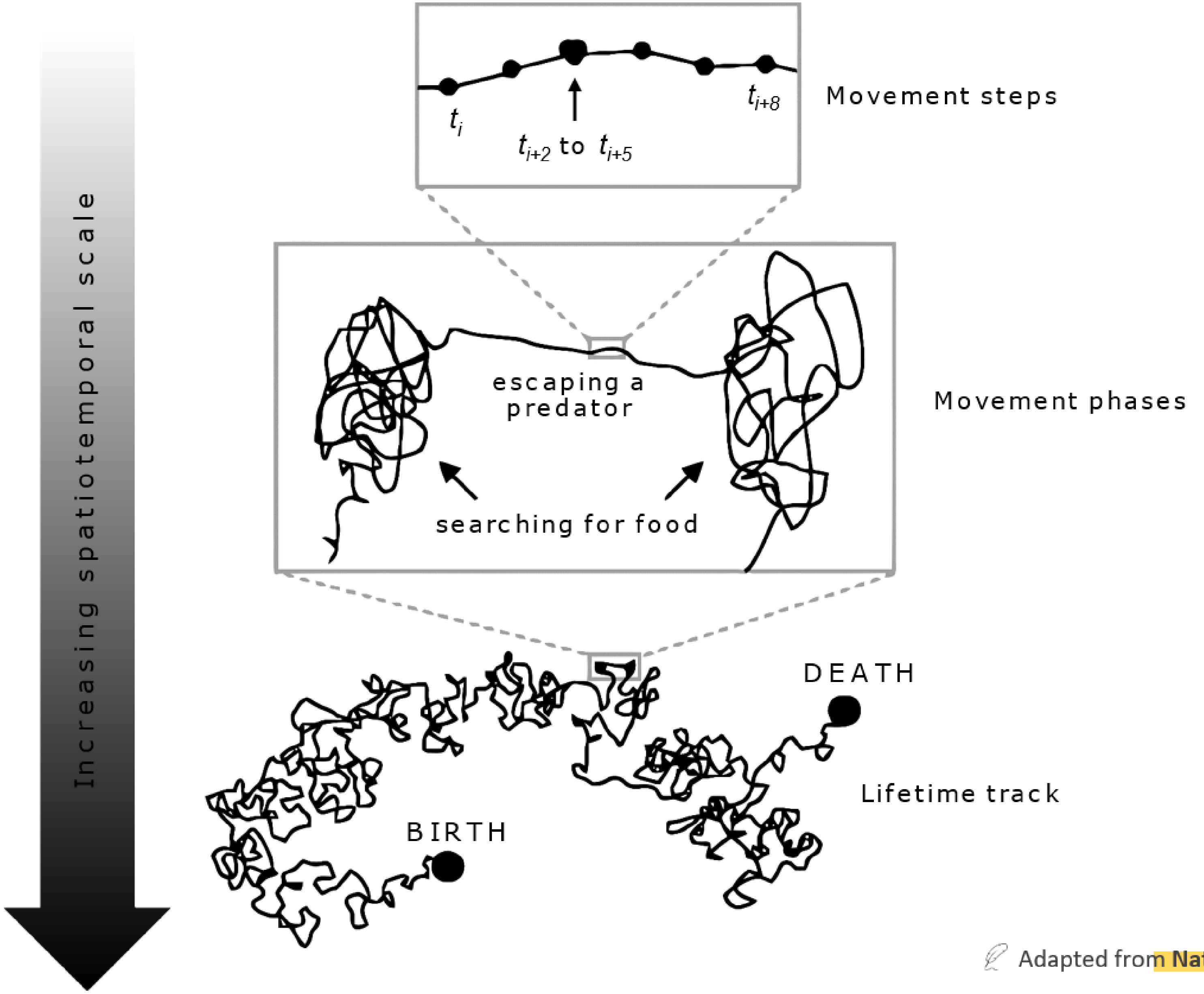
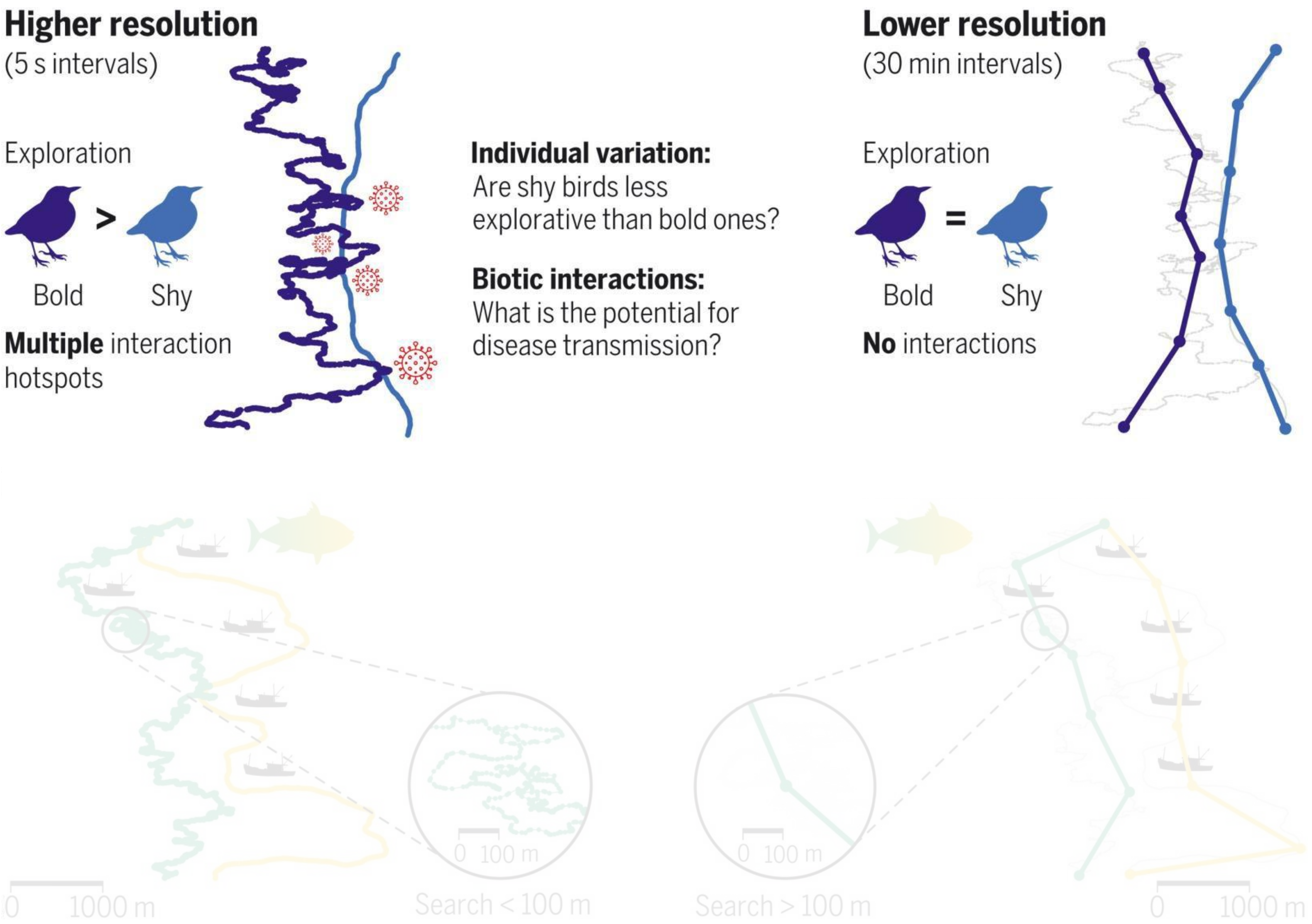


Adapted from **Nathan *et al.* (2008)**



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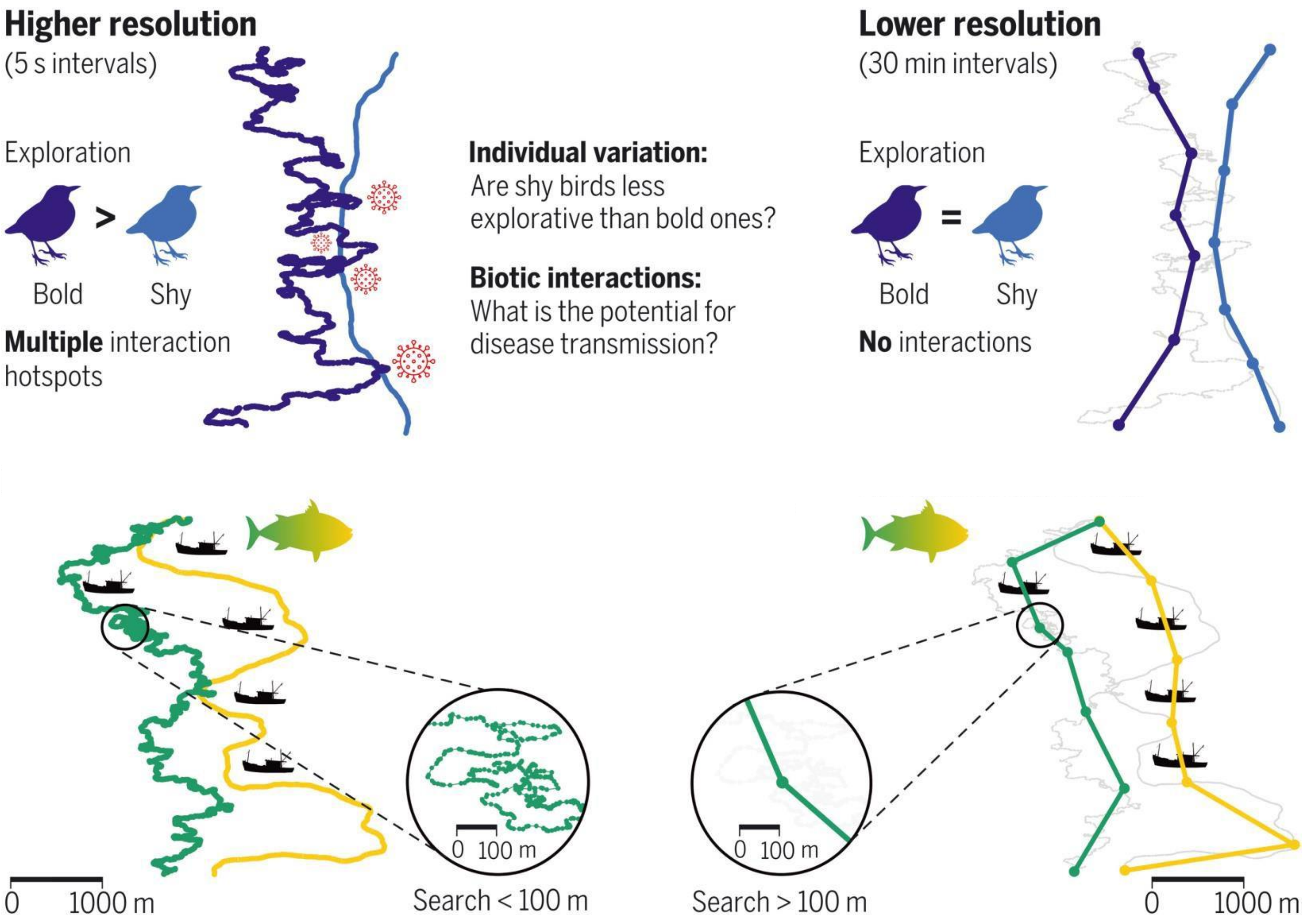

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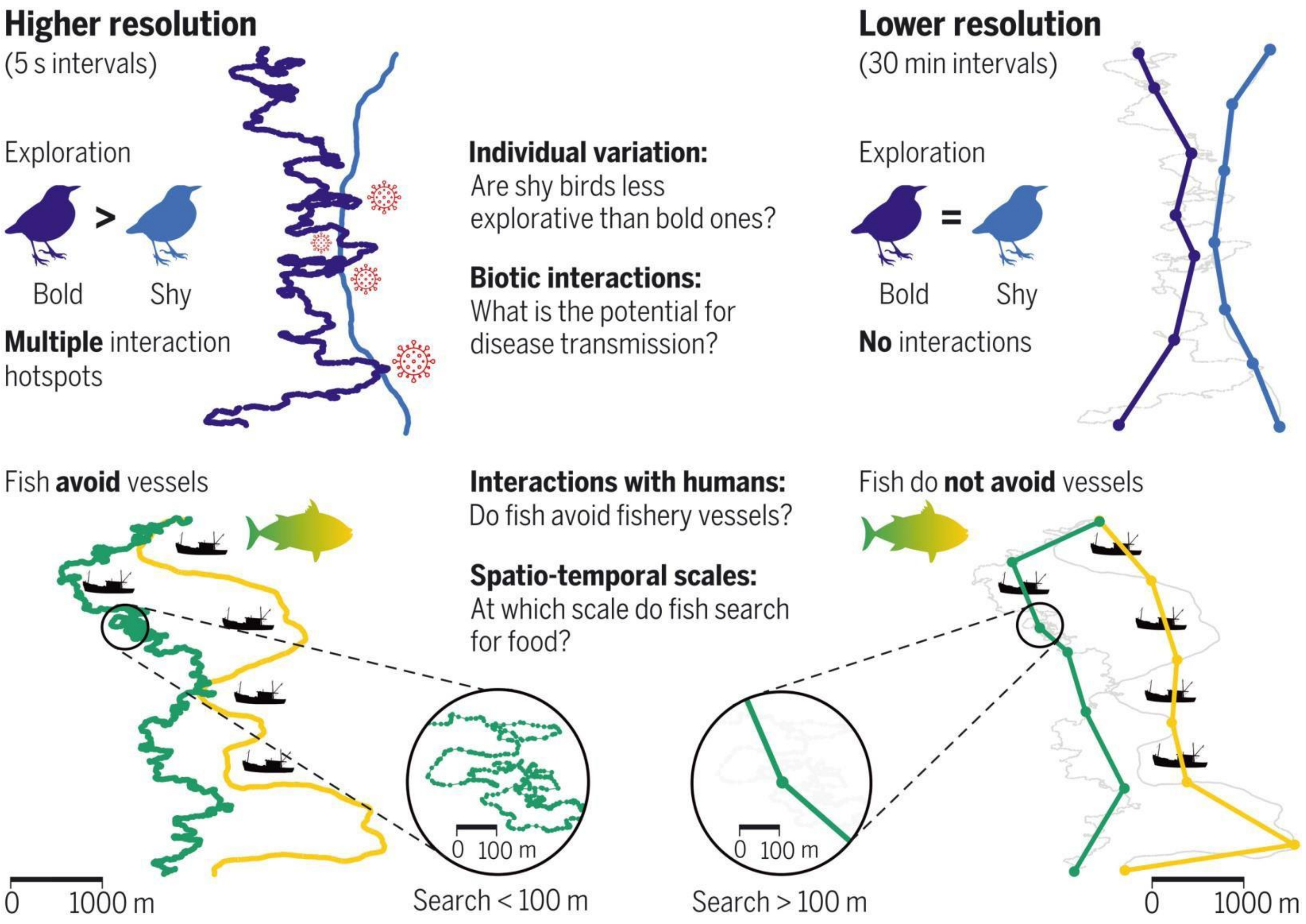

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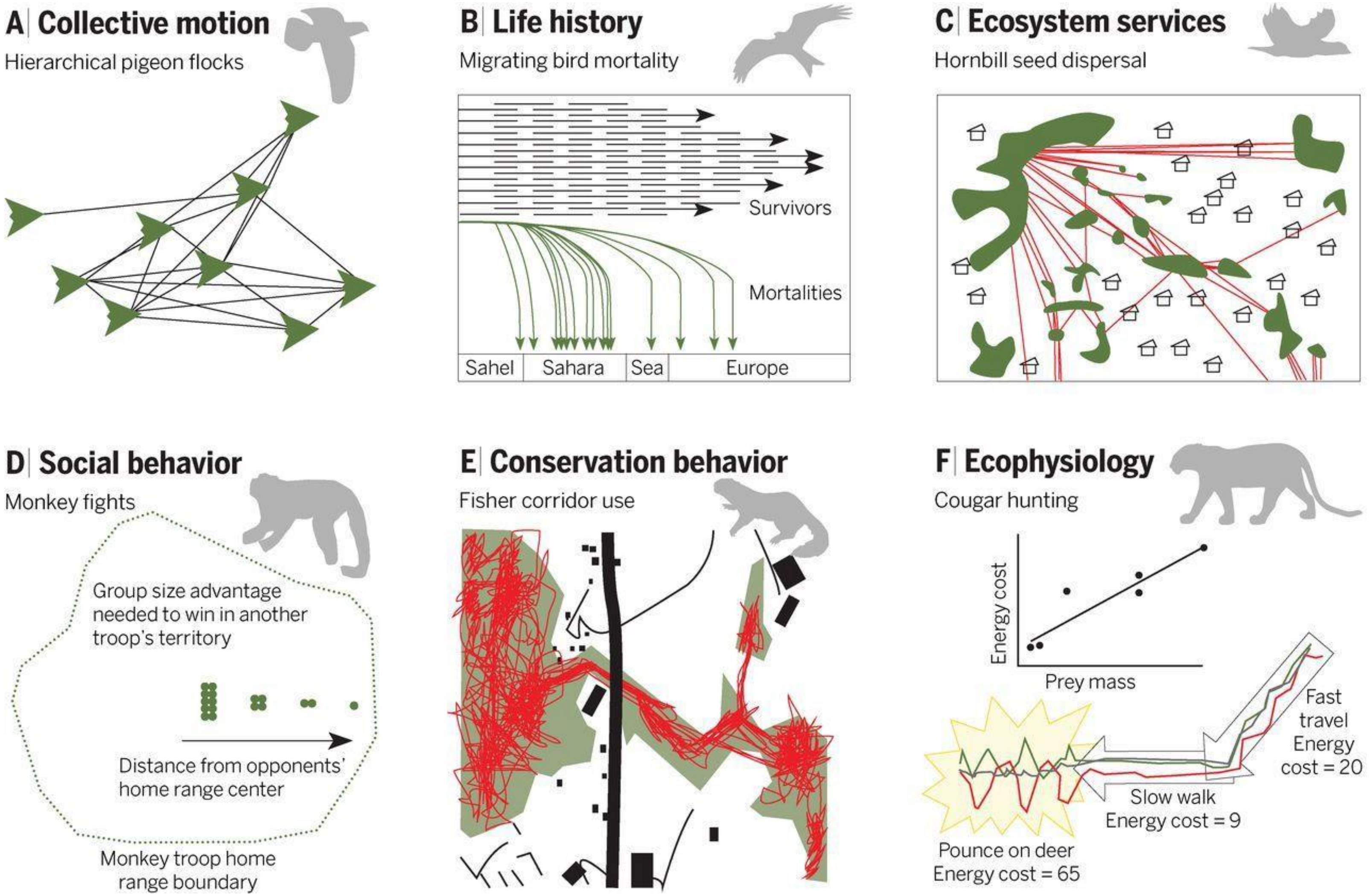

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DOI: 10.1126/science.aaa2478

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- Design **reintroduction strategies** and monitor their success

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- Evaluate the impact of **climate and land use change**
- Predict and prevent **human-wildlife conflict**
- Design **reintroduction strategies** and monitor their success
- Understand how **diseases spread** in wildlife populations

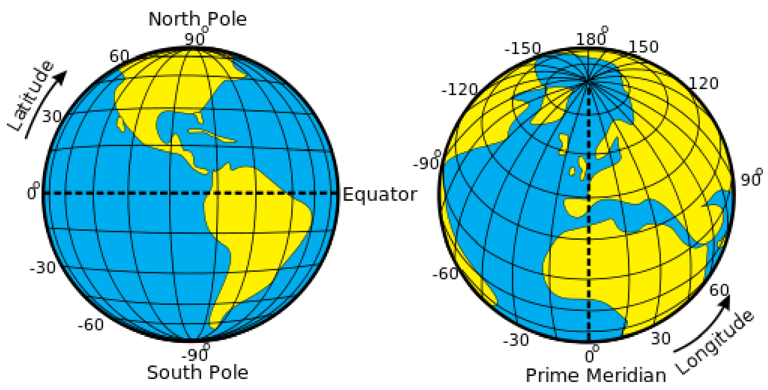
Overview

1) Introduction to movement ecology
2) **Projections**
3) Measurements
4) Orientation
5) Stochastic processes
6) Defining movement

Projections

To work with spatial data accurately, we need
Coordinate Reference Systems (CRS)

- **Geographic**
  - In which spherical coordinates are measured from the *Earth's center*
- **Planimetric**
  - Earth's coordinates are projected onto a *two-dimensional planar surface*



Geographic CRS (latitude/longitude)

Projected CRS (UTMs)

Projections

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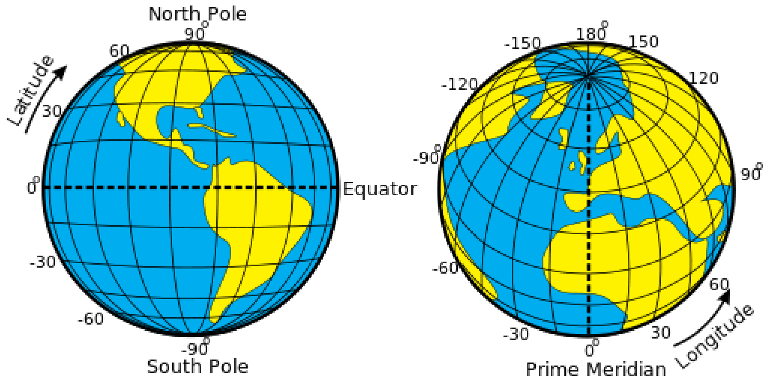


Geographic CRS (latitude/longitude)

- **Planimetric**
  - Earth's coordinates are projected onto a *two-dimensional planar surface*

Requires a *map projection*

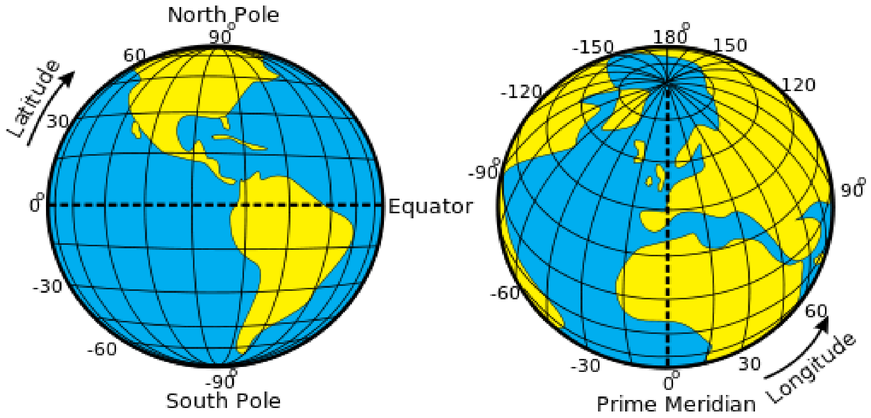
Measuring distances is easier, but the trade-off is that this only works over ***relatively small areas***.

Projected CRS (UTMs)

Projections

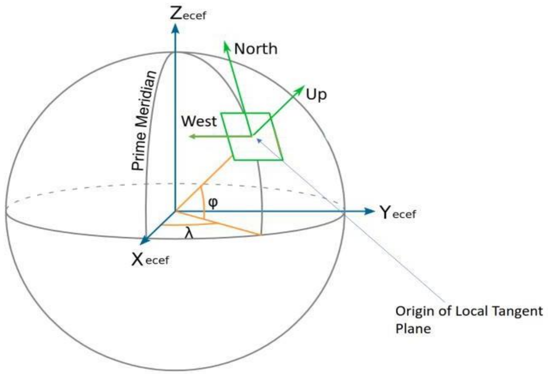
- **Geographic**
  - **EPSG:4326 (WGS84)**

+proj=longlat +elips=WGS84 +datum=WGS84 +no_defs



- **Planimetric**
  - **EPSG:32718 (UTM Zone 18S)**

+proj=utm +zone=18 +datum=WGS84 +elips=WGS84 +towgs84=0,0,0 +units=m +no_defs



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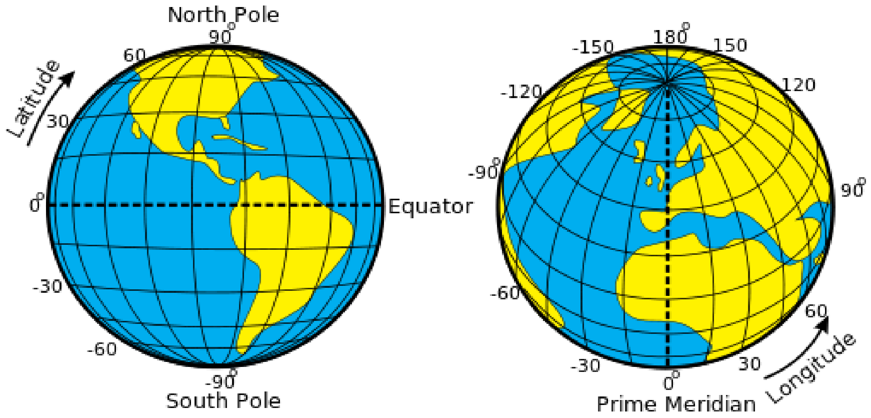

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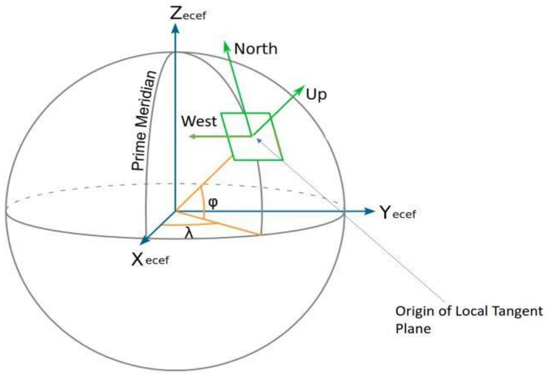

Ellipsoid + Datum:
Uses the WGS84 (World Geodetic System 1984) reference ellipsoid to approximate the Earth's shape (angular units and starting point, *e.g.*, 0,0).

UTM projections are typically built on top of WGS84 to maintain ***consistency***.

Projections

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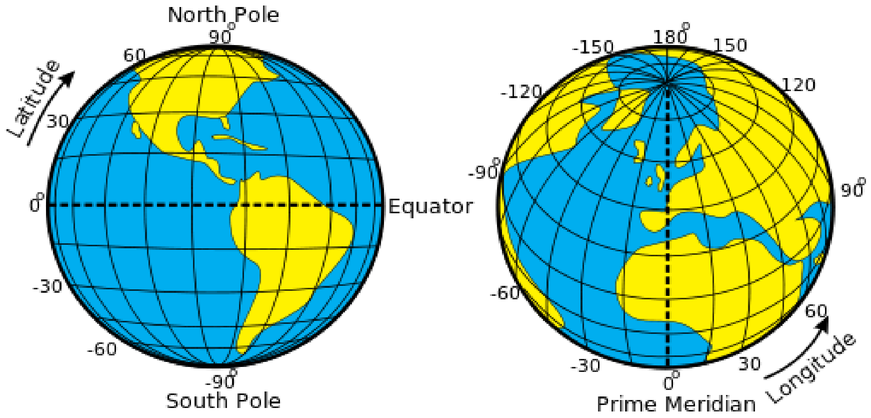

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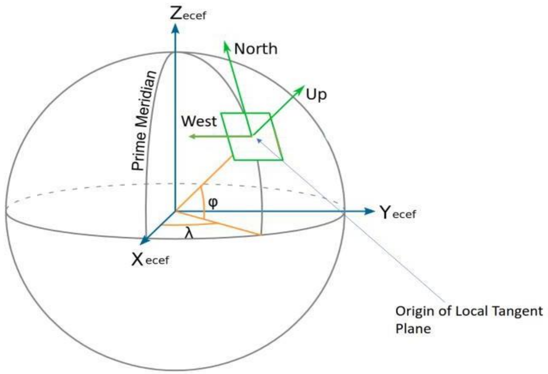

Specifies that no transformation is needed between this system and the WGS84 datum (no translation/rotation in X, Y, Z).

Specifies that coordinates are in **meters**.

Projections

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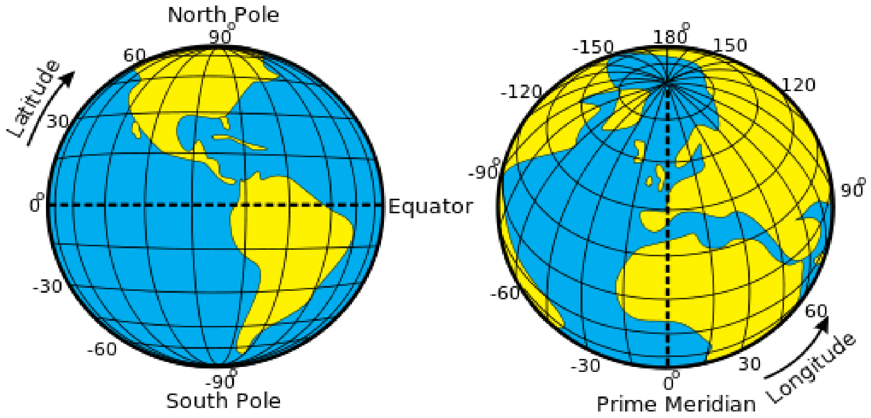
+proj=longlat +elips=WGS84 +datum=WGS84 **+no_defs**



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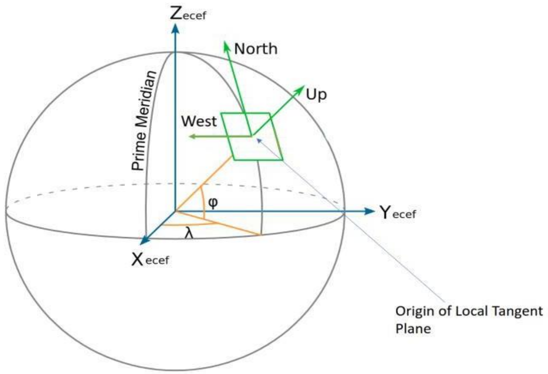

Ensures no additional default parameters are used in interpreting the projection.

Overview

1) Introduction to movement ecology
2) Projections
3) **Measurements**
4) Orientation
5) Stochastic processes
6) Autocorrelation and spectral density
7) Defining movement

Measurements

Vectors

- Specified by both magnitude (value + unit) and direction
    - Displacement
    - Velocity
    - Acceleration
    - N/A

Scalars

- Specified by only magnitude (value + unit)
    - Distance
    - Speed
    - Acceleration/deceleration
    - Time

Measurements

Vectors

- Specified by both *magnitude* (value + unit) and *direction*

Scalars

- Specified by only *magnitude* (value + unit)

magnitude of **displacement** vector

- **Displacement**
- Velocity
- Acceleration
- N/A

- **Distance**
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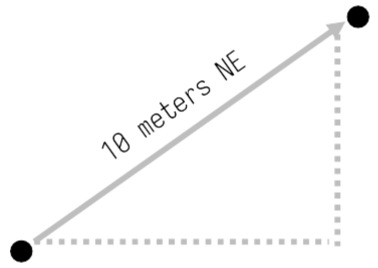

Magnitude: length of the arrow
Direction: direction of the arrow

10 meters

Measurements

Vectors

- Specified by both *magnitude* (value + unit) and *direction*

    - Displacement
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    - Acceleration
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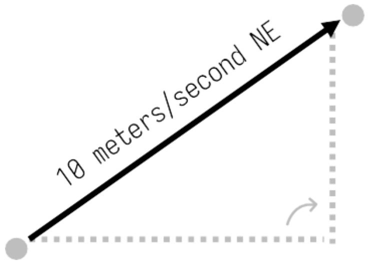
magnitude of **velocity** vector



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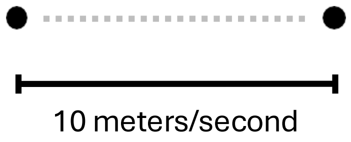

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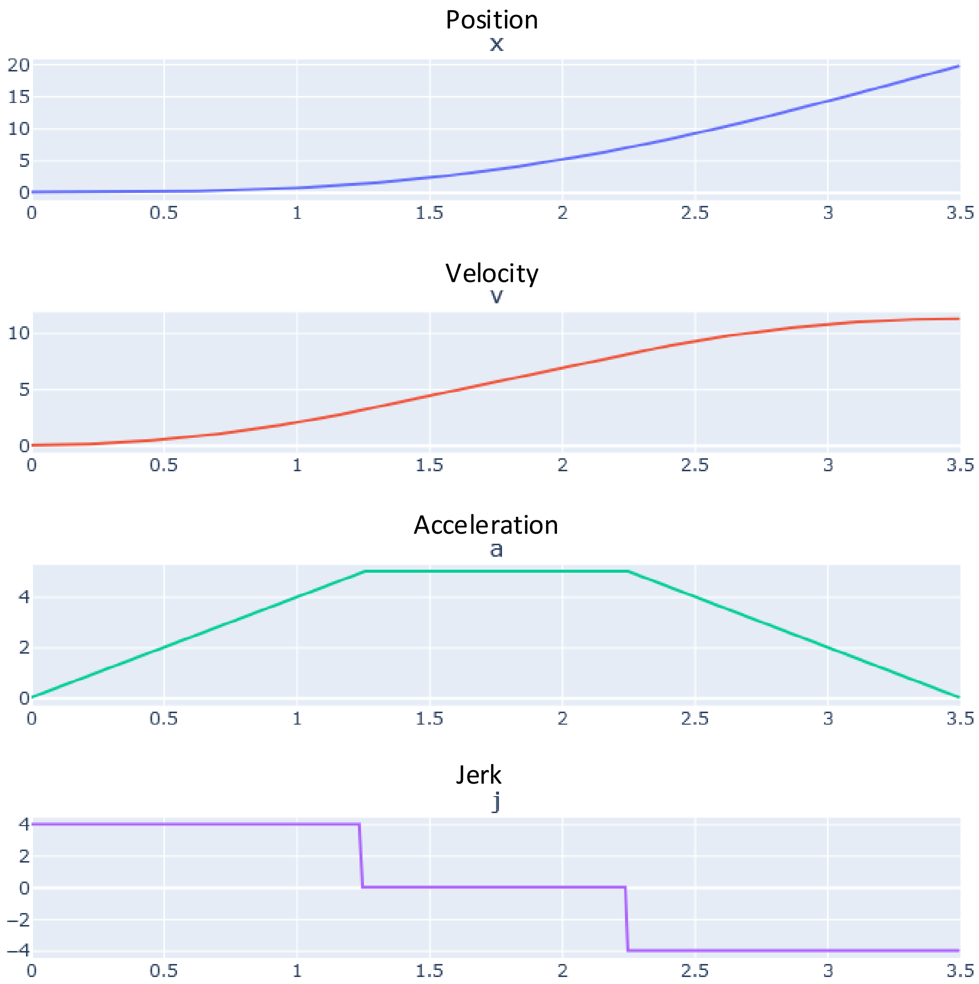


Fig. One-dimensional trajectory.

When an animal changes direction, the *first derivative* of its position is velocity. The *second derivative* is acceleration.

$\mathbf{r}$	Position
$\frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$	Velocity
$\frac{d^2\mathbf{r}}{dt^2} = \ddot{\mathbf{r}}$	Acceleration
$\frac{d^3\mathbf{r}}{dt^3} = \dddot{\mathbf{r}}$	Jerk

Measurements

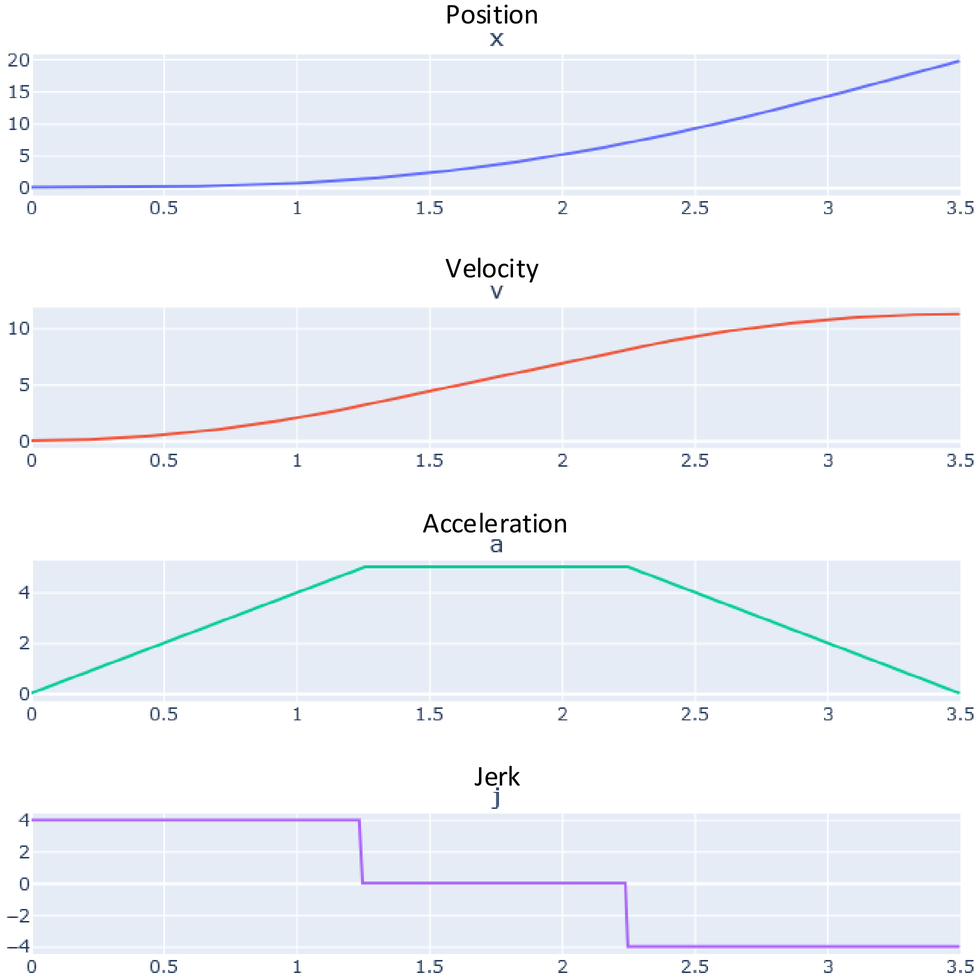


Fig. One-dimensional trajectory.

How fast and in what direction position changes

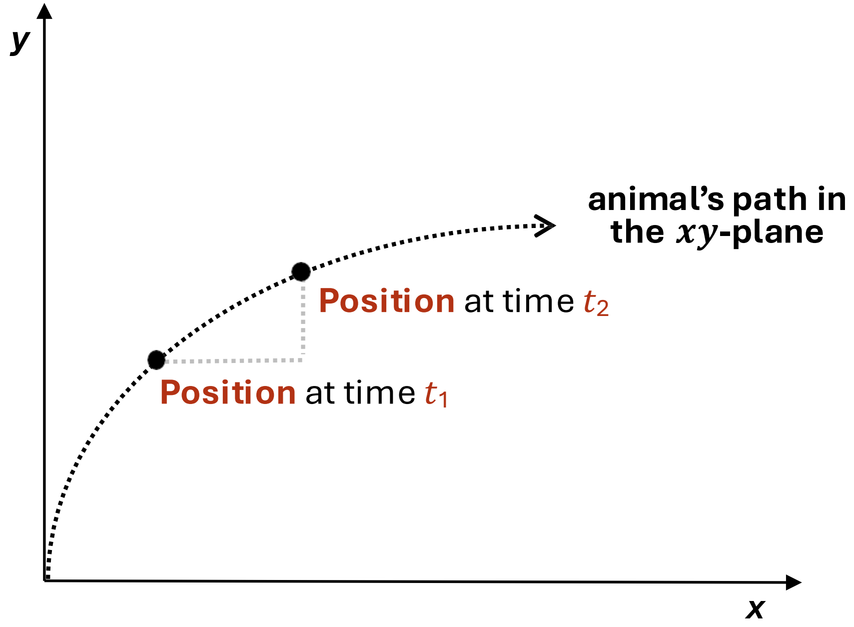
Speeding up or slowing down

How quickly acceleration itself changes over time

Measurements

- Average **velocity**:
  - *Displacement* divided by *time*
- Instantaneous **velocity**:
  - The instantaneous rate of change of the *position vector* with respect to *time*
  - (i.e., examined for a *very short time interval*)

$$V_t = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t}$$



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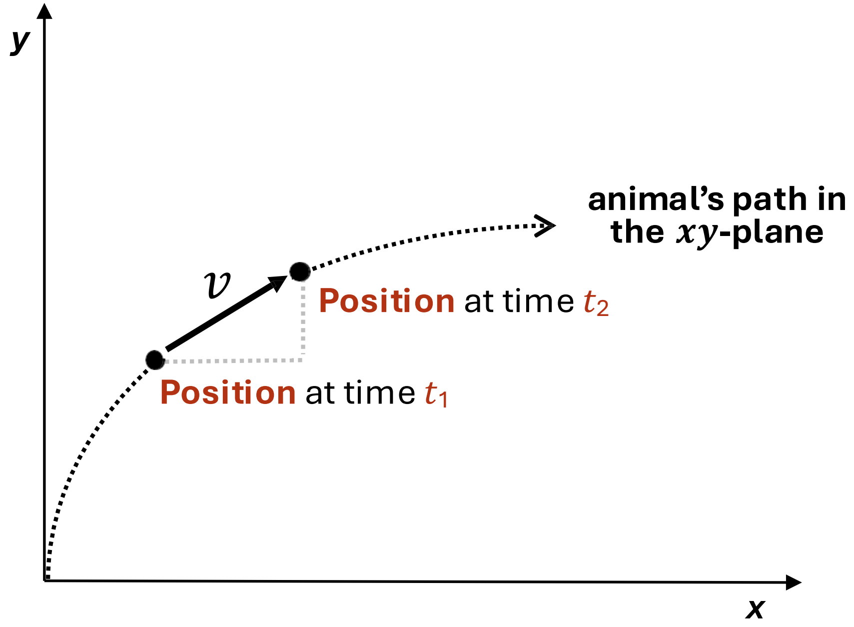

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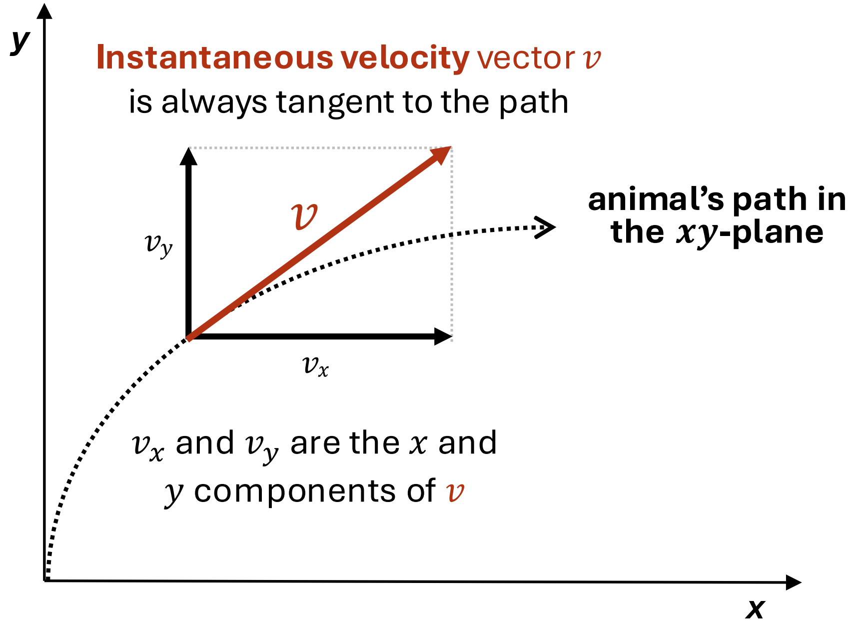

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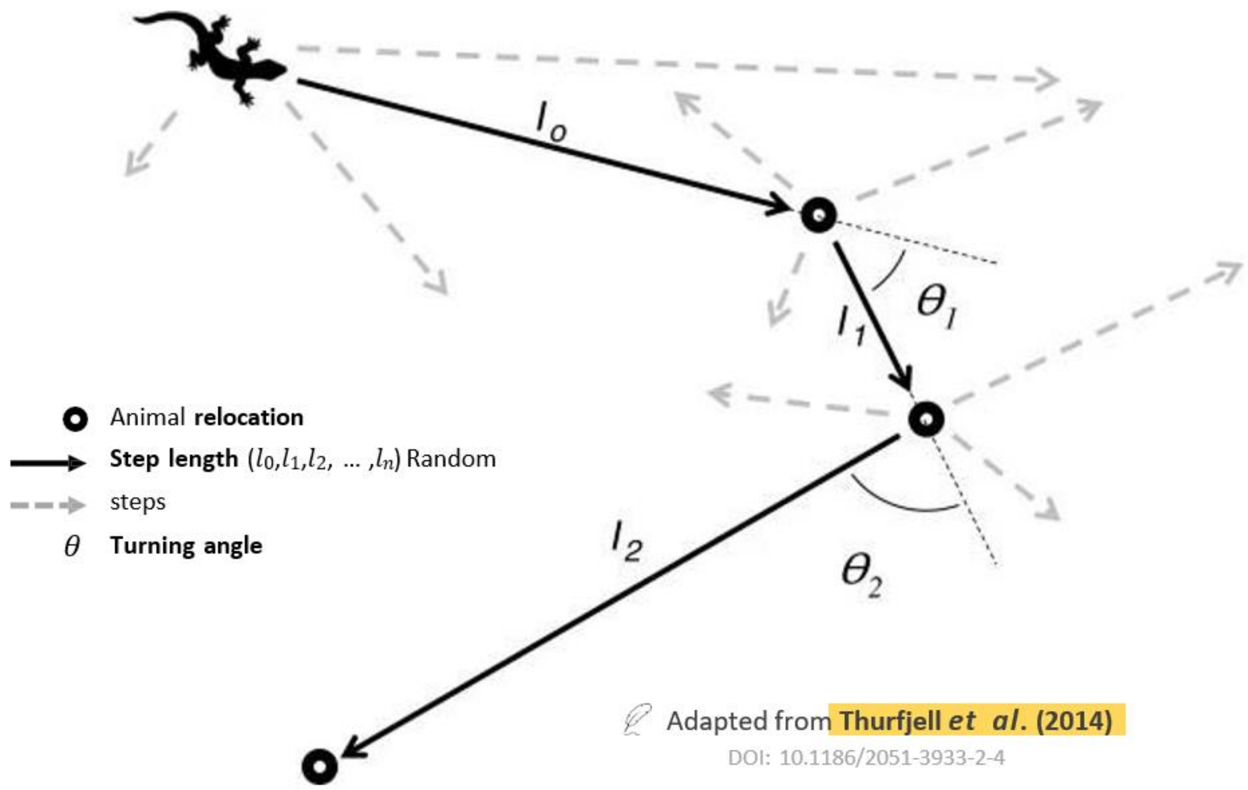

Adapted from **Thurfjell *et al.* (2014)**

DOI: 10.1186/2051-3933-2-4

Step **length**:

- Euclidean *distance* between two consecutive locations

Turning **angle**:

- Relative *internal* angles between consecutive locations, indicating change in *direction*

Overview

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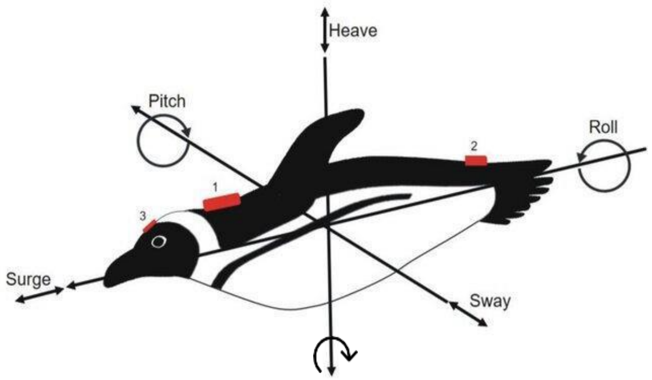
Orientation

- Accelerometers measure the linear acceleration of a body on each axis, which includes *heave*, *sway*, and *surge*



Del Caño *et al.* (2021)

DOI: 10.1186/s40462-022-00322-9

- **Translations**: linear motion along X, Y, and Z axes
  - Heave, sway, surge
- **Rotations**: angular motion around the X, Y, and Z axes
  - *Pitch*, *roll*, *yaw*
- **Heading**: direction where the nose of the animal is pointing
  - Usually expressed in degrees from North
- **Bearing**: angle of the animal relative to some point

Orientation

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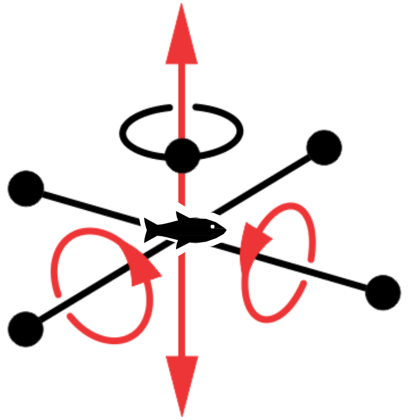
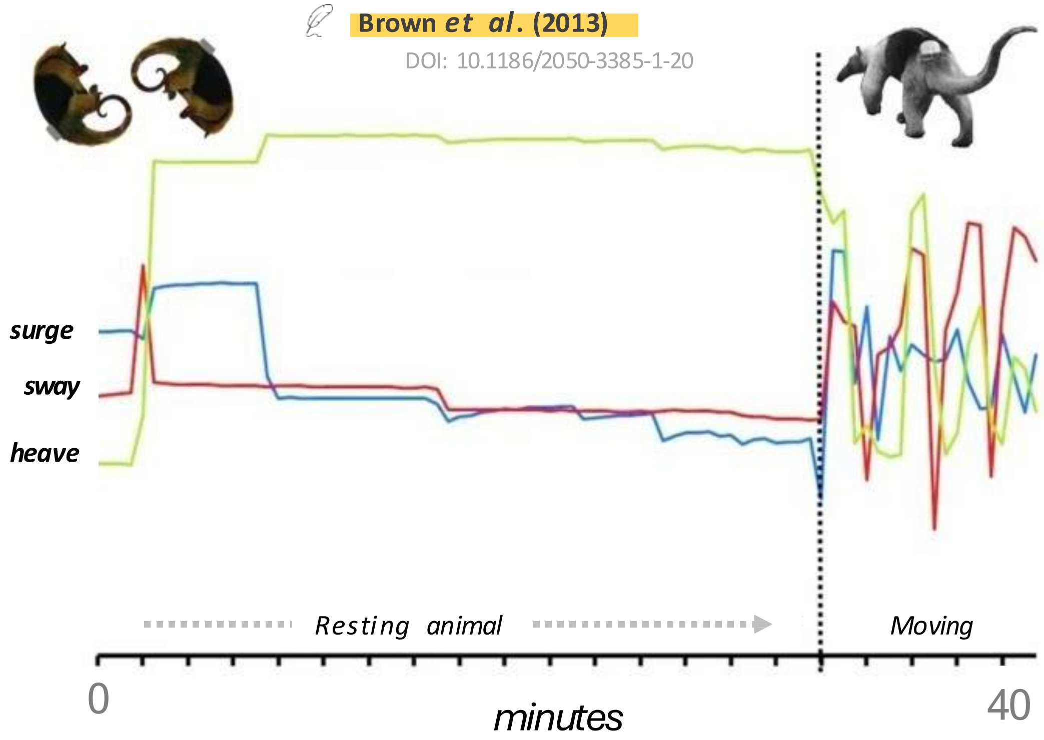


Intrinsic degrees of freedom
(mechanical DOFs, not statistical DOFs!)
=
the number of unique ways in which an animal can move in an alignment system.



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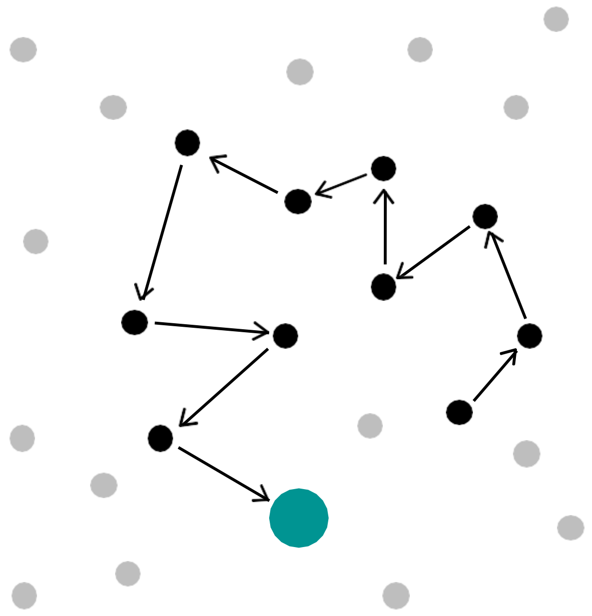
Stochastic processes

- A **stochastic process** describes the evolution of some phenomenon over time
  - "*Stochastic*" is generally interchangeable with "*random*," indicating an element of chance/unpredictability



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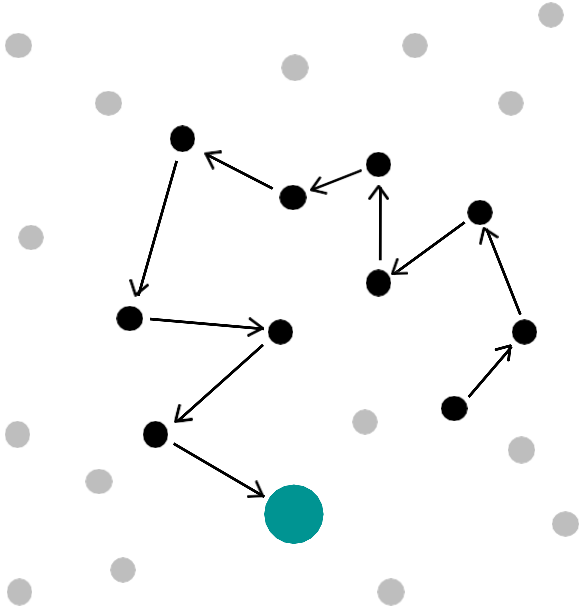

Does not imply that we have no information about the dynamics; a stochastic process provides structure with some ***randomness***.

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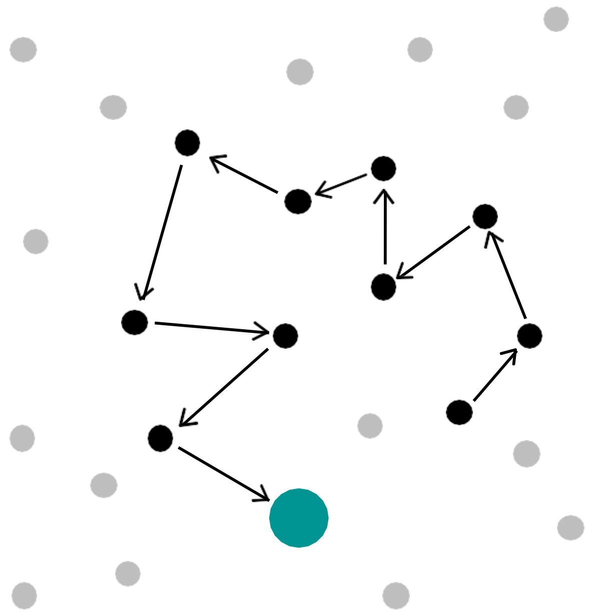


Markov process

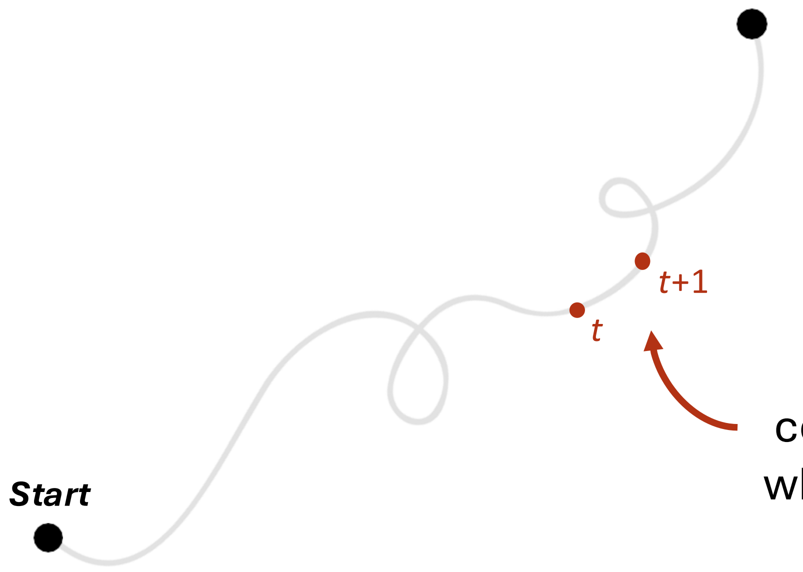
Type of stochastic process where the future state of the system depends *only on its current state*, not on the sequence of events that preceded it.

Brownian motion

Particles move *randomly*: Markov process where the next location is normally distributed around the current location with its variance proportional to how long you wait.

Movement

- An individual's spatial locations over time, forming a continuous trajectory

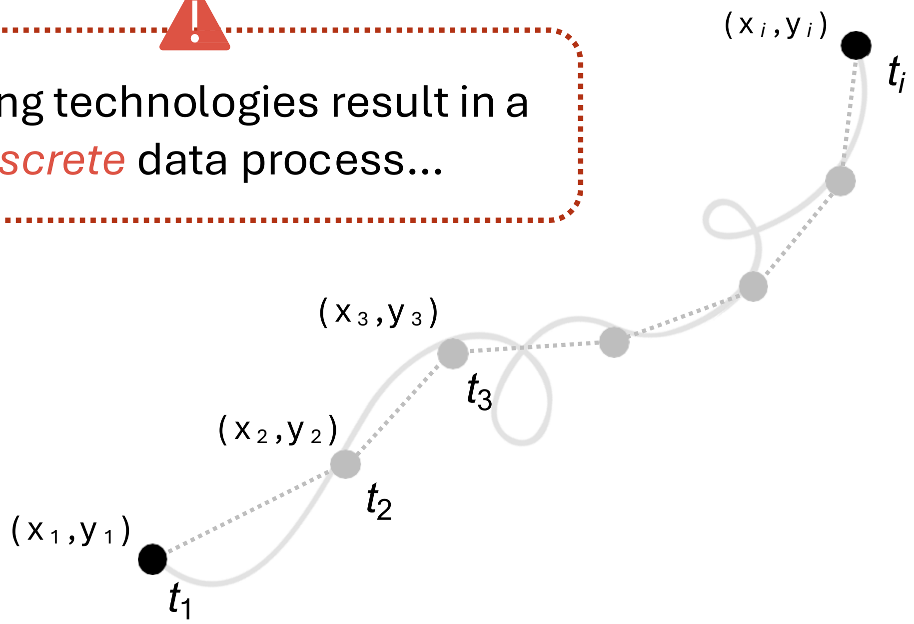


What happens at time *t* contains *information* about what will happen at time *t*+1

Movement

- An individual's spatial locations over time, forming a continuous trajectory

Tracking technologies result in a *discrete* data process...



Movement

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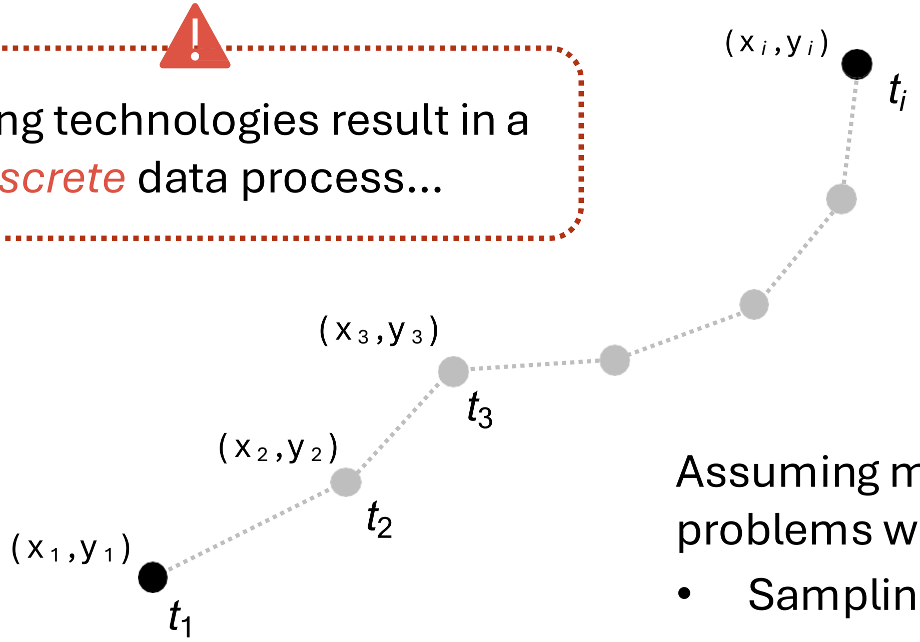
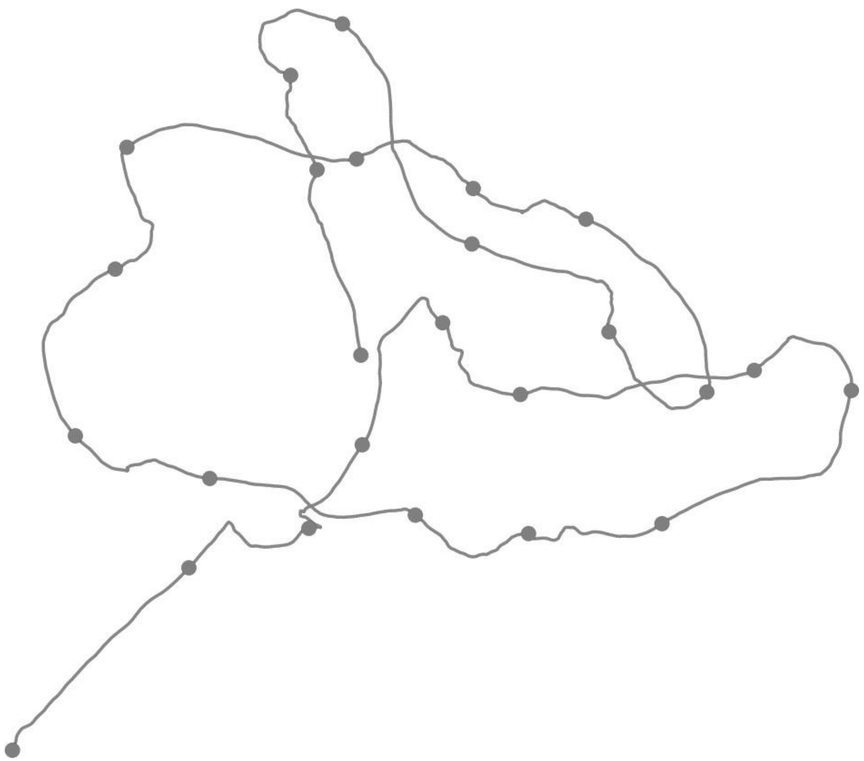


Assuming movement is also *discrete-time* may cause problems when:

- Sampling is *irregular* in time
- Sampling *schedules differ* across individuals.

Movement

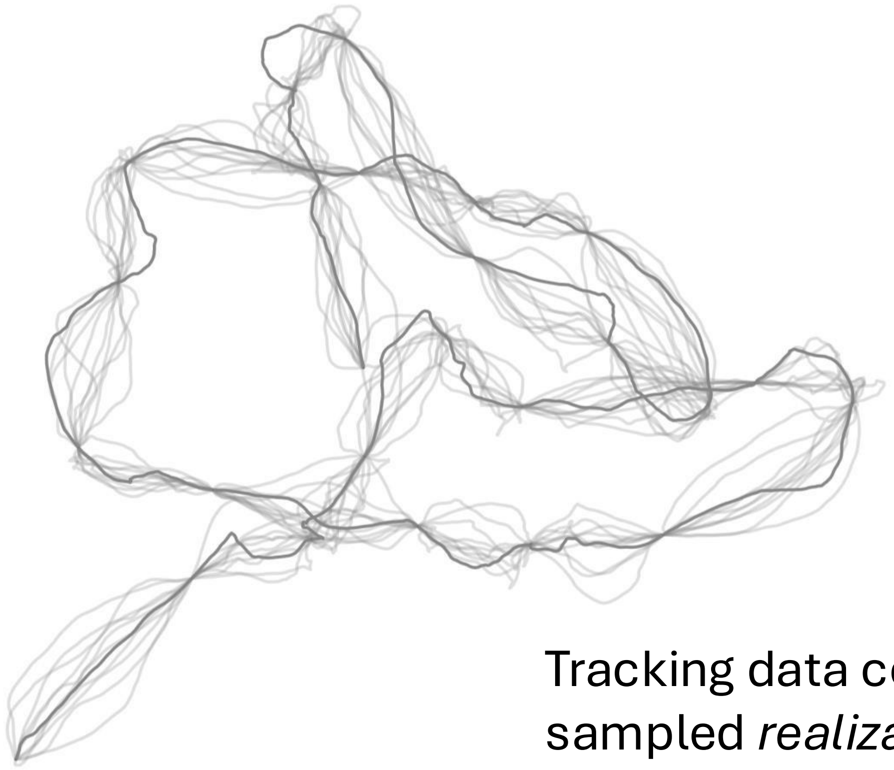
- How can we accurately characterize animal movement?
  - i.e., location, velocity, and direction over a period of time



Movement

- How can we accurately characterize animal movement?
  - i.e., location, velocity, and direction over a period of time

Continuous-time stochastic process



Tracking data correspond to one sparsely sampled *realization* of the movement process.

Thank you

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